BLG562E: Parallel Computing For GPUs using CUDA

Spring 2026

Week 4- CUDA Programming

Section Outline

- CPU + GPU Heterogeneous Computing
- Simple CUDA code examples
  - Hello World
  - addition(), vector_addition
- Memory Management
- Cooperating Threads
- Managing The Device

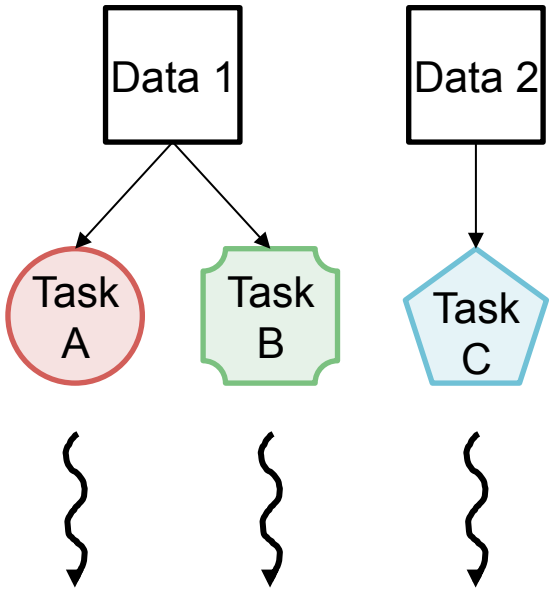
Types of Parallelism

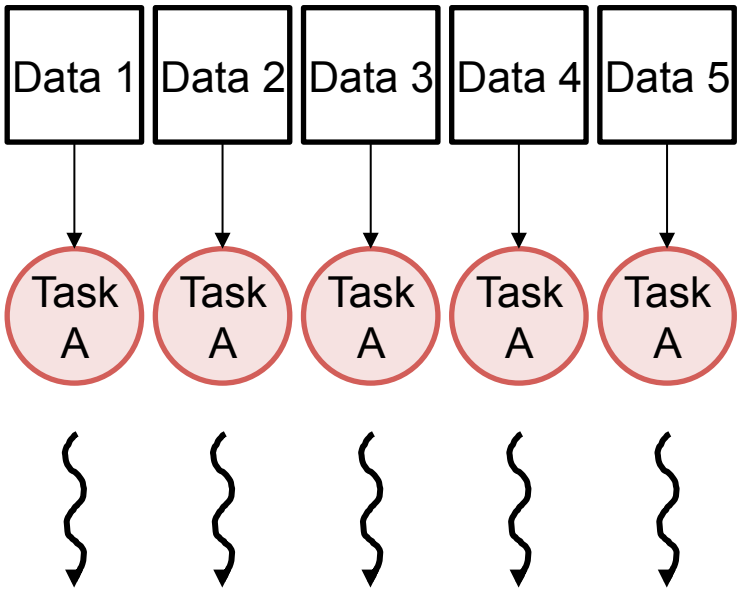
Task Parallelism

- Different operations performed on same or different data
- Usually a modest number of tasks unleashing a modest amount of parallelism

Data Parallelism

- Same operations performed on different data
- Potentially massive amounts of data unleashing massive amounts of parallelism
  - Most suitable for GPUs





GPUs: A Highly Multithreaded Coprocessor

- The GPU is viewed as a compute **device** that:
  - Is a coprocessor to the CPU or **host**
  - Has its own DRAM (**device memory**)
  - Runs many **threads in parallel**

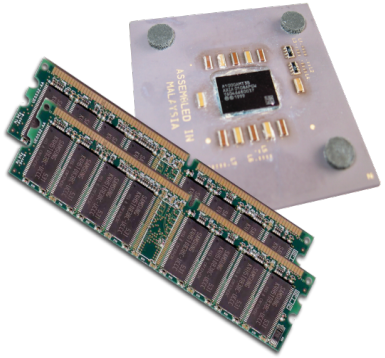
- Data-parallel portions of an application are executed on the device as **kernels** which run in parallel on many threads

GPUs: A Highly Multithreaded Coprocessor

- Differences between GPU and CPU threads
  - GPU threads are extremely lightweight
    - Very little creation overhead
  - GPU needs 1000s of threads for full efficiency
    - Multi-core CPU needs only a few

CPU + GPU Heterogeneous Computing

- ***Host*** The CPU and its memory (host memory)
- ***Device*** The GPU and its memory (device memory)



Host

Device

CPU + GPU Heterogeneous Computing

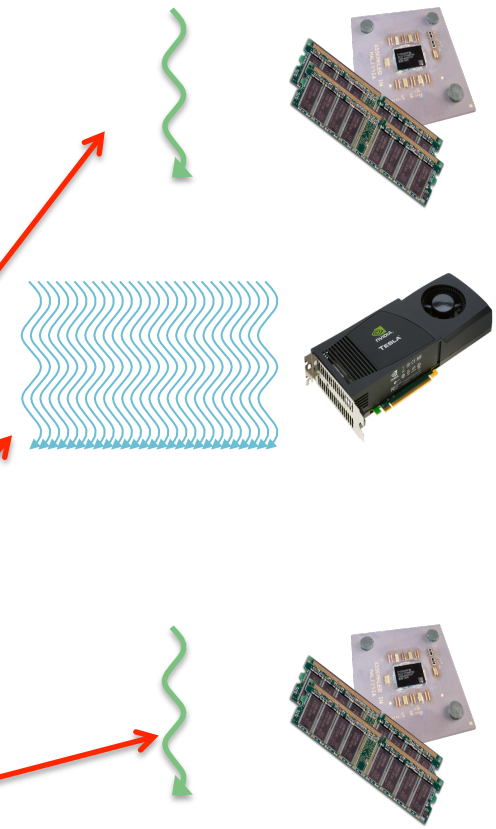
```
#include <iostream>
#include <algorithm>

using namespace std;

#define N         1024
#define RADIUS    3
#define BLOCK_SIZE 16

__global__ void stencil_1d(int *in, int *out) {
        __shared__ int temp[BLOCK_SIZE + 2 * RADIUS];
        int gindex = threadIdx.x + blockIdx.x * blockDim.x;
        int lindex = threadIdx.x + RADIUS;

        // Read input elements into shared memory
        temp[lindex] = in[gindex];
        if (threadIdx.x < RADIUS) {
                temp[lindex - RADIUS] = in[gindex - RADIUS];
                temp[lindex + BLOCK_SIZE] = in[gindex + BLOCK_SIZE];
        }

        // Synchronize (ensure all the data is available)
        __syncthreads();

        // Apply the stencil
        int result = 0;
        for (int offset = -RADIUS ; offset <= RADIUS ; offset++)
                result += temp[lindex + offset];

        // Store the result
        out[gindex] = result;
}

void fill_ints(int *x, int n) {
        fill_n(x, n, 1);
}

int main(void) {
        int *in, *out;          // host copies of a, b, c
        int *d_in, *d_out;      // device copies of a, b, c
        int size = (N + 2*RADIUS) * sizeof(int);

        // Alloc space for host copies and setup values
        in  = (int *)malloc(size); fill_ints(in,  N + 2*RADIUS);
        out = (int *)malloc(size); fill_ints(out, N + 2*RADIUS);

        // Alloc space for device copies
        cudaMalloc((void **)&d_in,  size);
        cudaMalloc((void **)&d_out, size);

        // Copy to device
        cudaMemcpy(d_in,  in,  size, cudaMemcpyHostToDevice);
        cudaMemcpy(d_out, out, size, cudaMemcpyHostToDevice);

        // Launch stencil_1d() kernel on GPU
        stencil_1d<<<N/BLOCK_SIZE,BLOCK_SIZE>>>(d_in + RADIUS, d_out + RADIUS);
        // Copy result back to host
        cudaMemcpy(out, d_out, size, cudaMemcpyDeviceToHost);

        // Cleanup
        free(in); free(out);
        cudaFree(d_in); cudaFree(d_out);
        return 0;
}
```

parallel fn

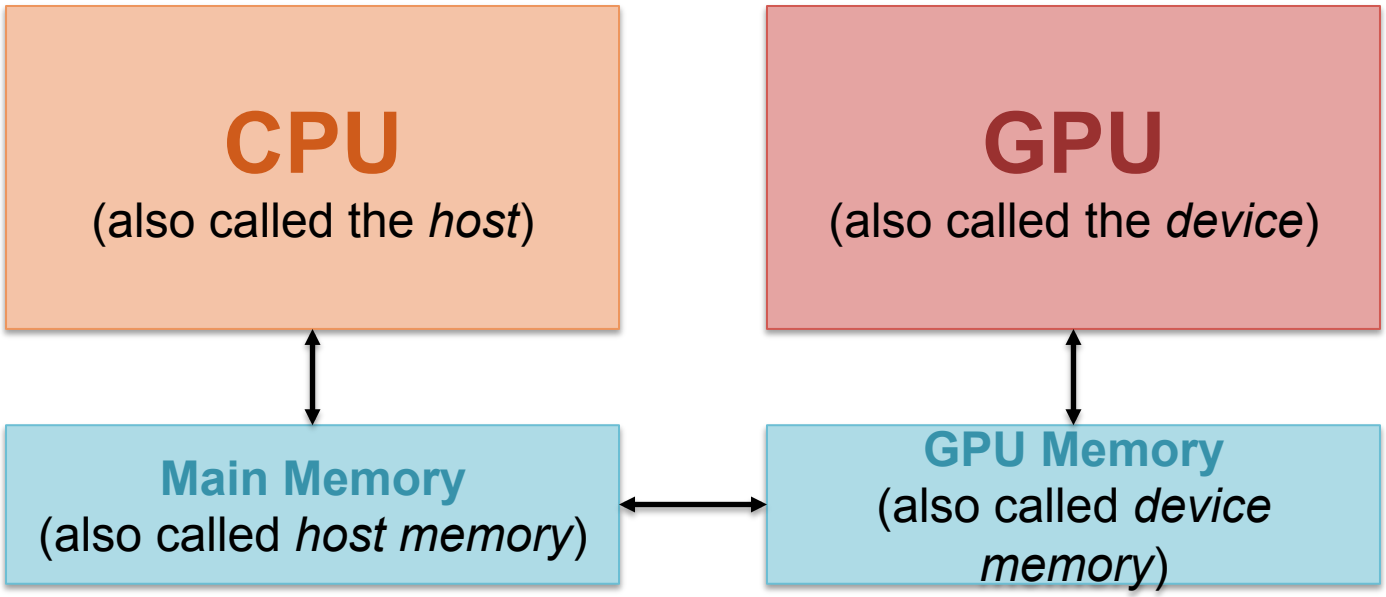
serial code

parallel code

serial code



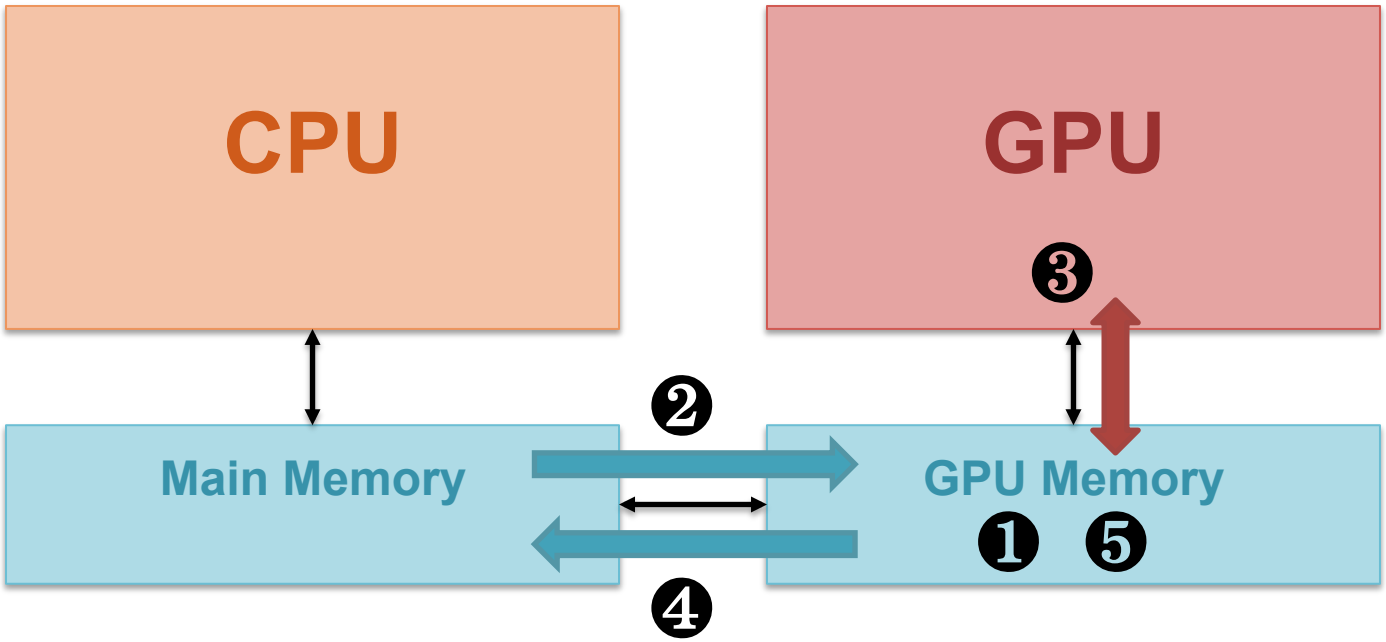
System Organization



The CPU and GPU have separate memories can cannot access each others' memories

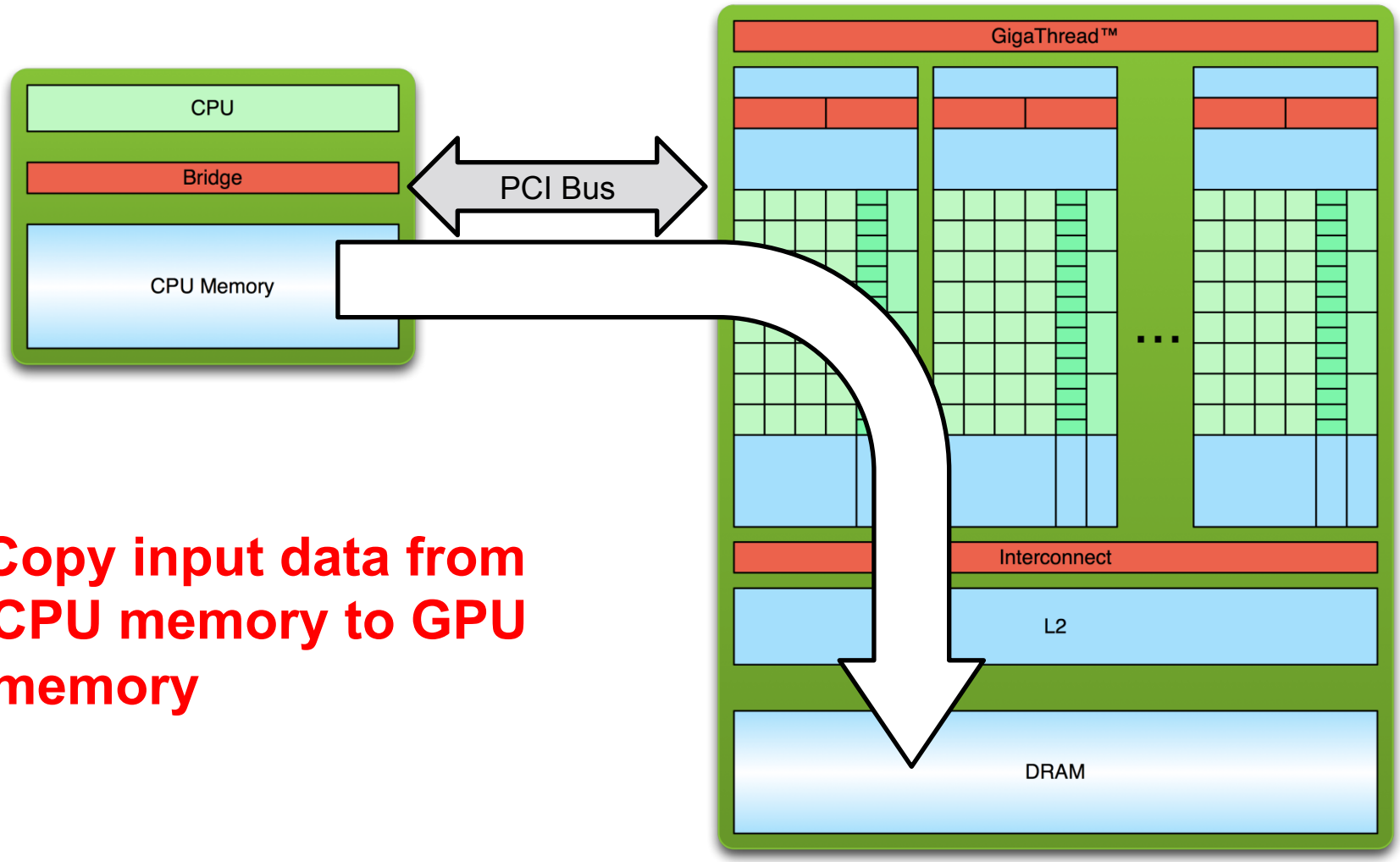
Need to transfer data between them

Memory Management



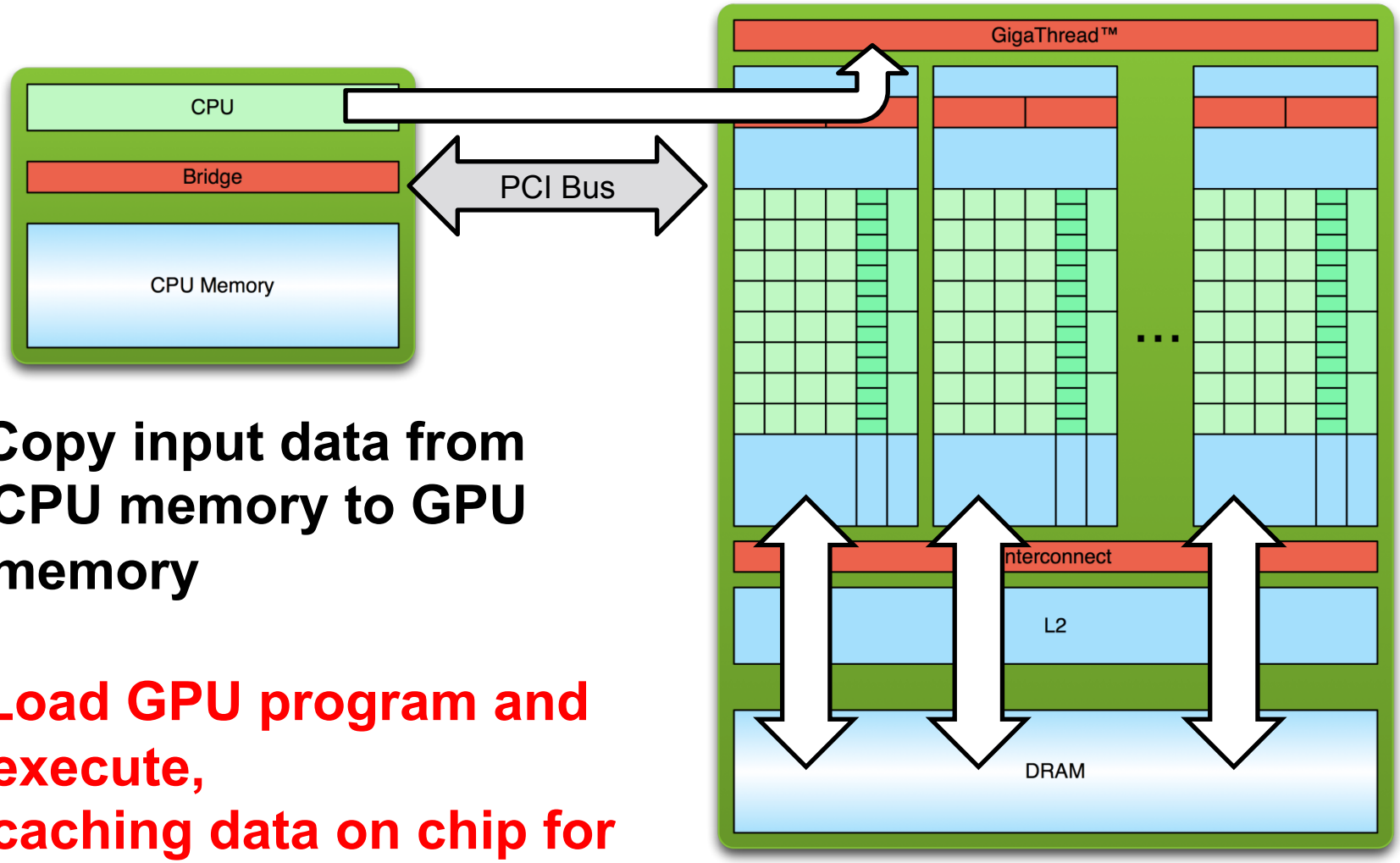
❶ Allocate GPU memory
❷ Copy data to GPU memory
❸ Perform computation on GPU
❹ Copy data from GPU memory
❺ Deallocate GPU memory

Simple Processing Flow



1. Copy input data from CPU memory to GPU memory

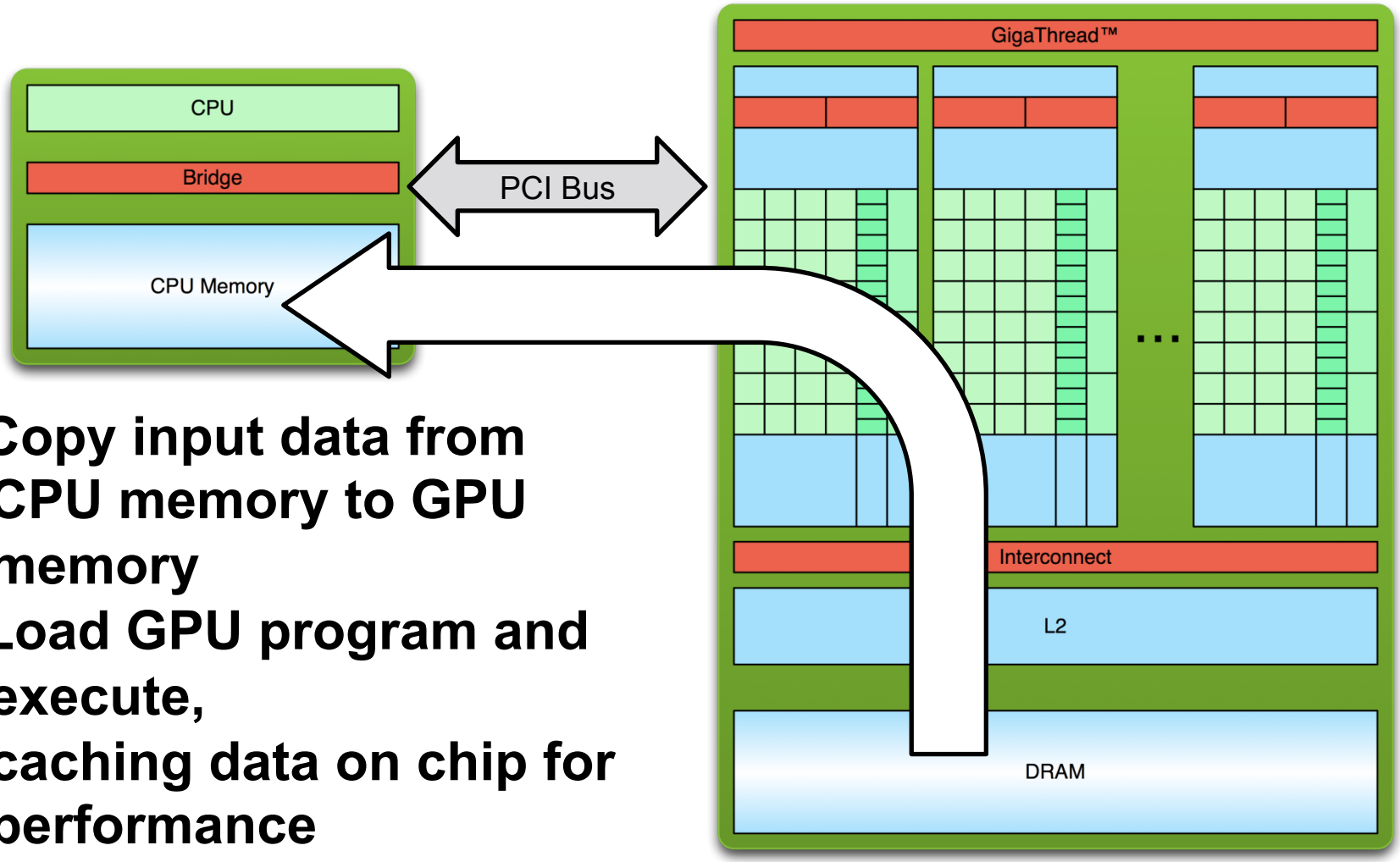
Simple Processing Flow



1. Copy input data from CPU memory to GPU memory

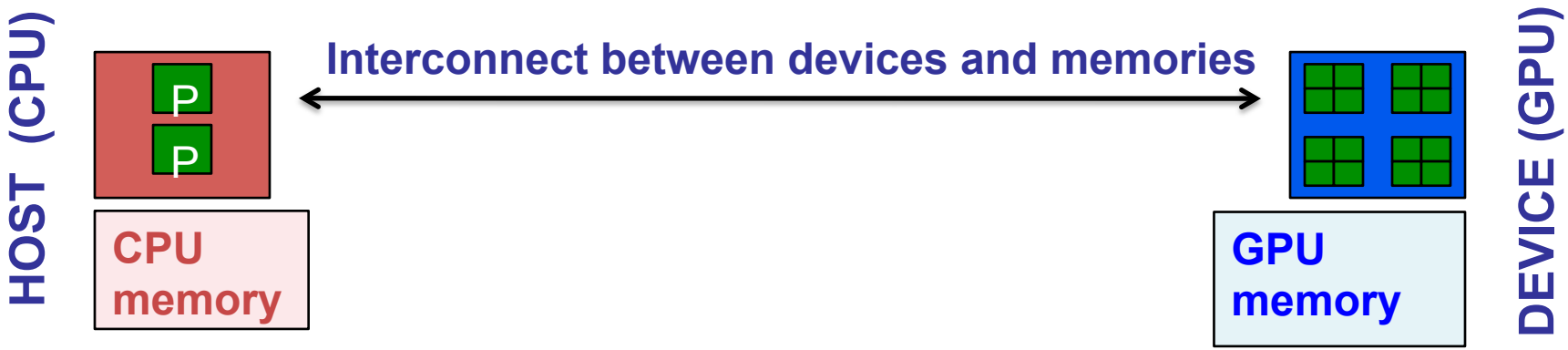
2. Load GPU program and execute,
   caching data on chip for performance

Simple Processing Flow



1. Copy input data from CPU memory to GPU memory
2. Load GPU program and execute,
   caching data on chip for performance
3. Copy results from GPU memory to CPU memory

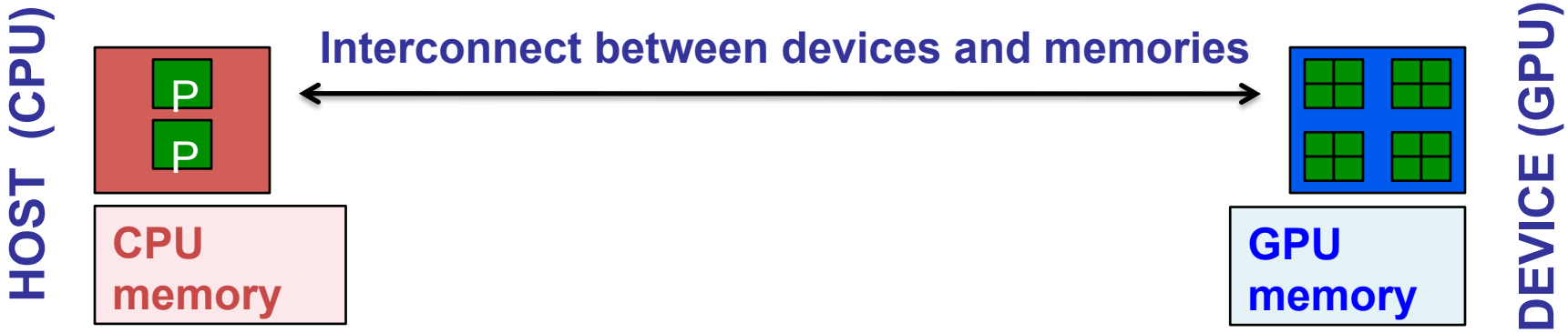
What Programmer Expresses in CUDA



- Computation partitioning (where does computation occur?)
  - Declarations of functions __host__, __global__, __device__
  - Mapping of thread programs to device:

```
compute <<<gs, bs>>>(<args>)
```

What Programmer Expresses in CUDA



- Data partitioning (where does data reside, who may access it and how?)
  - Declarations of data __shared__, __device__, __constant__, …
- Data management and orchestration
  - Copying to/from host:

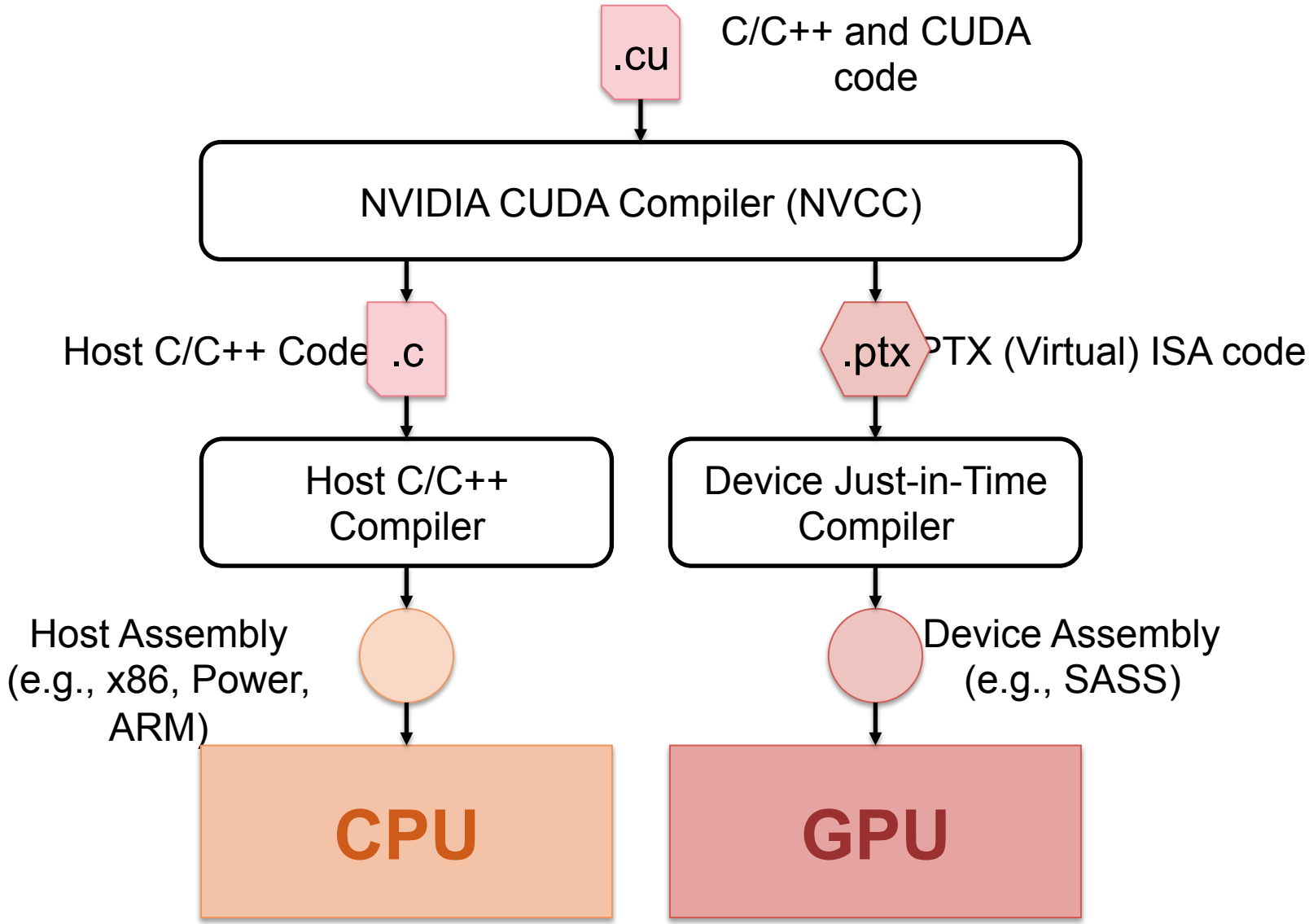
    e.g., cudaMemcpy(h_obj,d_obj, cudaMemcpyDevicetoHost)
- Concurrency management
  - *E.g.* __synchthreads()

Minimal Extensions to C + API

- **Declspecs**
  - global, device, shared, local, constant
- **Keywords**
  - threadIdx, blockIdx
- **Intrinsics**
  - __syncthreads
- **Runtime API**
  - Memory, symbol, execution management
- **Function launch**

```
__device__ float filter[N];

__global__ void convolve (float *image)
{

  __shared__ float region[M];
  ...

region[threadIdx] = image[i];

  __syncthreads()
  ...

  image[j] = result;
}

// Allocate GPU memory
void *myimage = cudaMalloc(bytes)


// 100 blocks, 10 threads per block
convolve<<<100, 10>>> (myimage);
```

Compilation



NVCC Compiler's Role: Partition Code and Compile for Device

mycode.cu

```
int main_data;
__shared__  int sdata;

Main() { }
__host__  hfunc () {
  int hdata;
 <<<gfunc(g,b,m)>>>();
}

__global__ gfunc() {
  int gdata;
}

__device__ dfunc() {
  int ddata;
}
```

Host Only

Interface

Device Only

Compiled by native compiler: gcc, icc, cc

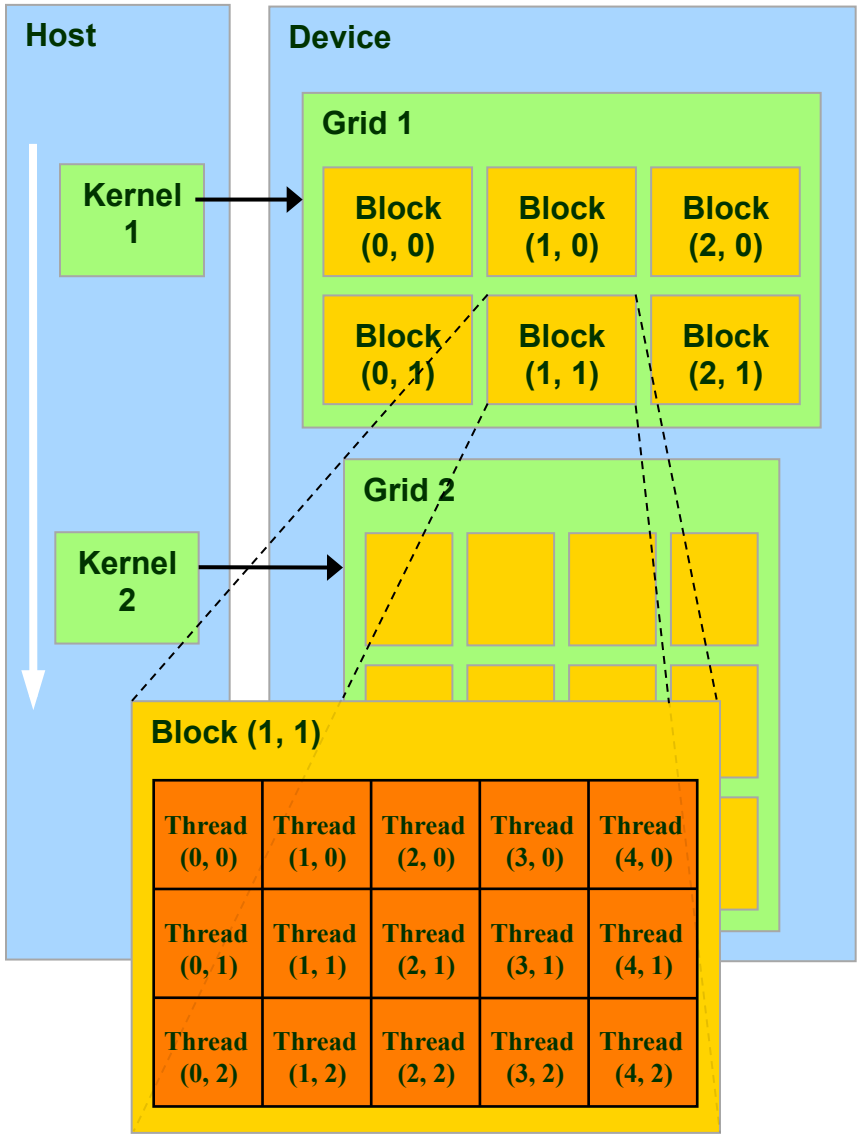
```
int main_data;

Main() {}
__host__  hfunc () {
  int hdata;
<<<gfunc(g,b,m)>>>
();
}
```

Compiled by nvcc compiler

```
__shared__ sdata;

__global__ gfunc() {
  int gdata;
}

__device__ dfunc() {
  int ddata;
}
```

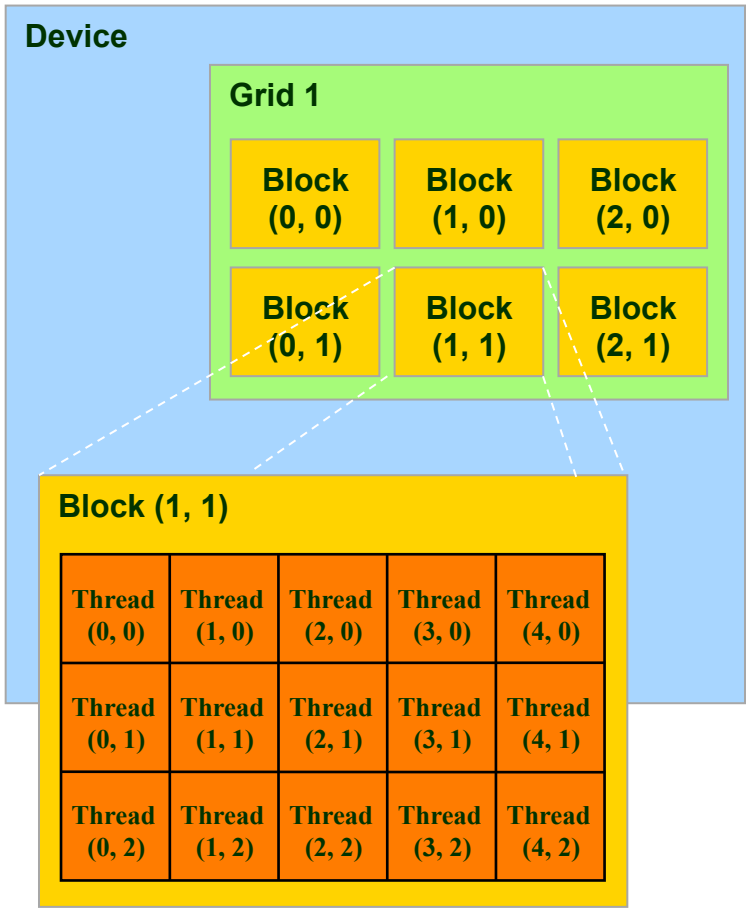
Thread Batching: Grids and Blocks

- A kernel is executed as a grid of thread blocks
  - All threads share data memory space
- A thread block is a batch of threads that can cooperate with each other by:
  - Synchronizing their execution
    - For hazard-free shared memory accesses
  - Efficiently sharing data through a low latency shared memory
- Two threads from two different blocks cannot cooperate



Block and Thread IDs

- Threads and blocks have IDs
  - So each thread can decide what data to work on
  - Block ID: 1D or 2D (blockIdx.x, blockIdx.y)
  - Thread ID: 1D, 2D, or 3D (threadIdx.{x,y,z})
- Simplifies memory addressing when processing multidimensional data
  - Image processing
  - Solving PDEs on volumes
  - …



First CUDA Code Example: Hello World!

```
int main(void) {
        printf("Hello World!\n");
        return 0;
}
```

- Standard C that runs on the host
- NVIDIA compiler (nvcc) can be used to compile programs with no *device* code

```
$ nvcc -o gpu_hello gpu_hello.cu
$ ./gpu_hello
Hello World!
$
```

Hello World! with Device Code

```
__global__ void mykernel(void)
{
}

int main(void) {
      mykernel<<<1,1>>>();
      printf("Hello World!\n");
      return 0;
}
```

Hello World! with Device Code

```
__global__ void mykernel(void)
{
}
```

- CUDA C/C++ keyword `__global__` indicates a function that:
  - Runs on the device
  - Is called from host code

- `nvcc` separates source code into host and device components
  - Device functions (e.g. `mykernel()`) processed by NVIDIA compiler
  - Host functions (e.g. `main()`) processed by standard host compiler
    - `gcc, icc, cc, etc..`

Hello World! with Device Code

```
mykernel<<<1,1>>>();
```

- Triple angle brackets mark a call from *host* code to *device* code
  - Also called a “kernel launch”
  - We’ll get to the parameters (1,1) in a moment
- That’s all that is required to execute a function on the GPU!

Hello World! with Device Code

```
__global__ void mykernel(void)
{
}
int main(void) {
      mykernel<<<1,1>>>();
      printf("Hello World!\n");
      return 0;
}
```

Note: `mykernel()` does nothing in this example

```
$ nvcc -o gpu_hello gpu_hello.cu
$ ./gpu_hello
Hello World!
$
```

CUDA example with Computation on the Device

- A simple kernel to add two integers

```
__global__ void addition(int *a, int *b, int *c)
{
        *c = *a + *b;
}
```

- As before `__global__` is a CUDA C/C++ keyword meaning
  - `addition()` will execute on the device
  - `addition()` will be called from the host

Addition on the Device

- We use pointers for the variables

```
__global__ void addition(int *a, int *b, int *c)
{
        *c = *a + *b;
}
```

- `addition()` runs on the device, so `a`, `b` and `c` must point to device memory

- We need to allocate memory on the GPU

Memory Management

- Host and device memory are separate entities
  - ***Device*** pointers point to GPU memory

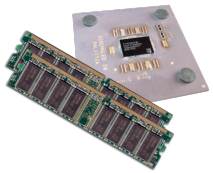
    May be passed to/from host code

    May *not* be dereferenced in host code

  - ***Host*** pointers point to CPU memory

    May be passed to/from device code

    May *not* be dereferenced in device code



- Simple CUDA API for handling device memory
  - **`cudaMalloc()`**, **`cudaFree()`**, **`cudaMemcpy()`**
  - Similar to the C equivalents `malloc()`, `free()`, `memcpy()`

CUDA Memory Management API

- Allocating memory:

```
cudaError_t cudaMalloc(void **devPtr, size_t
size)
```

  - `devPtr`: Pointer to pointer to allocated device memory
  - `size`: Requested allocation size in bytes

- Deallocating memory:

```
cudaError_t cudaFree(void *devPtr)
```

  - `devPtr`: Pointer to device memory to free

- Return type: `cudaError_t`
  - Helps with error checking (discussed later)

CUDA Memory Management API

- Copying memory:

```
cudaError_t cudaMemcpy(void *dst, const void
*src,
                size_t count, enum cudaMemcpyKind
kind)
```

  - `dst`: Destination memory address
  - `src`: Source memory address
  - `count`: Size in bytes to copy
  - `kind`: Type of transfer
    - `cudaMemcpyHostToHost`
    - `cudaMemcpyHostToDevice`
    - `cudaMemcpyDeviceToHost`
    - `cudaMemcpyDeviceToDevice`

Addition on the Device: `main()`

```
int main(void) {
        int a, b, c;                    // host copies of a, b, c
        int *d_a, *d_b, *d_c;           // device copies of a, b, c
        int size = sizeof(int);

        // Allocate space for device copies of a, b, c
        cudaMalloc((void **)&d_a, size);
        cudaMalloc((void **)&d_b, size);
        cudaMalloc((void **)&d_c, size);

        // Setup input values
        a = 2;
        b = 7;
```

Addition on the Device: `main()`

```
        // Copy inputs to device
        cudaMemcpy(d_a, &a, size, cudaMemcpyHostToDevice);
        cudaMemcpy(d_b, &b, size, cudaMemcpyHostToDevice);

        // Launch add() kernel on GPU
        addition<<<1,1>>>(d_a, d_b, d_c);

        // Copy result back to host
        cudaMemcpy(&c, d_c, size, cudaMemcpyDeviceToHost);

        // Cleanup
        cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);
        return 0;
}
```

Moving to Parallel

- GPU computing is about massive parallelism
  - So how do we run code in parallel on the device?

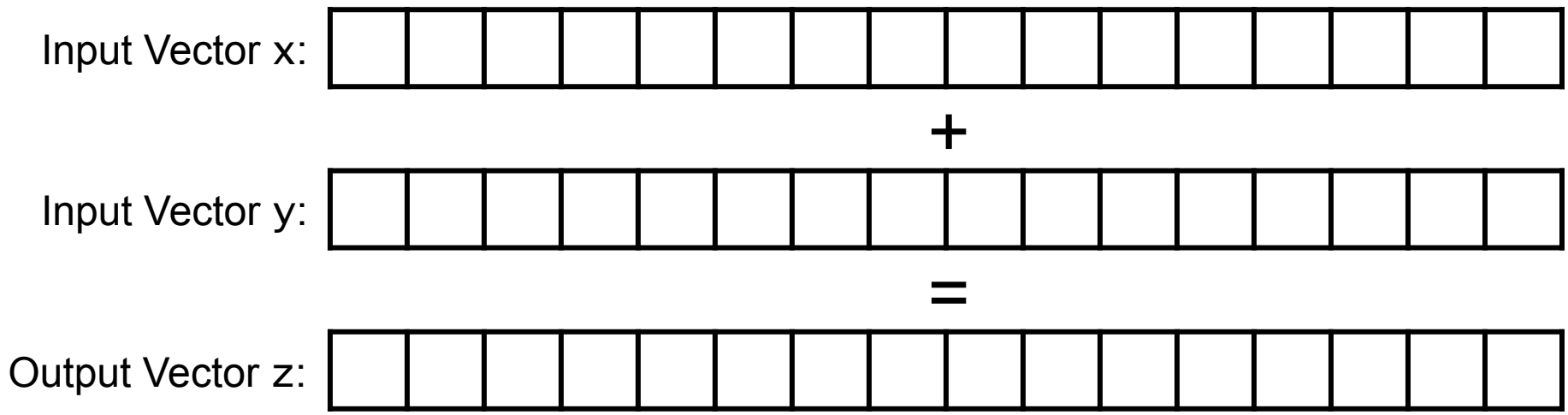
```
addition<<< 1, 1 >>>();
```

```
vector_addition<<< N, 1 >>>();
```

- Instead of executing `addition()` once, execute N times in parallel

Vector Addition

- Vector addition




```
void vecadd(float* x, float* y, float* z, int N) {
    for(unsigned int i = 0; i < N; ++i) {
        z[i] = x[i] + y[i];
    }
}
```

Sequential C Code

Vector Addition on the Device

- With `addition()` running in parallel we can do vector addition

- **Terminology:** each parallel invocation of `vector_addition()` is referred to as a ***block***
  - The set of blocks is referred to as a ***grid***
  - Each invocation can refer to its block index using `blockIdx.x`

```
__global__ void vector_addition(int *a, int *b, int *c) {
        c[blockIdx.x] = a[blockIdx.x] + b[blockIdx.x];
}
```

- By using ***`blockIdx.x`*** to index into the array, each block handles a different index

Vector Addition on the Device

```
__global__ void vector_addition(int *a, int *b, int *c) {
        c[blockIdx.x] = a[blockIdx.x] + b[blockIdx.x];
}
```

- On the device, each block can execute in parallel:

Block 0	Block 1	Block 2	Block 3
`c[0] = a[0] + b[0];`	`c[1] = a[1] + b[1];`	`c[2] = a[2] + b[2];`	`c[3] = a[3] + b[3];`

Vector Addition on the Device: `main()`

```
#define N 512
int main(void) {
   int *a, *b, *c              // host copies of a, b, c
   int *d_a, *d_b, *d_c;       // device copies of a, b, c
   int size = N * sizeof(int);

   // Alloc space for device copies of a, b, c
   cudaMalloc((void **)&d_a, size);
   cudaMalloc((void **)&d_b, size);
   cudaMalloc((void **)&d_c, size);

   // Alloc space for host copies of a, b, c
   //                              and setup input values
   a = (int *)malloc(size); random_ints(a, N);
   b = (int *)malloc(size); random_ints(b, N);
   c = (int *)malloc(size);
```

Vector Addition on the Device: `main()`

```
    // Copy inputs to device
    cudaMemcpy(d_a, a, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_b, b, size, cudaMemcpyHostToDevice);

    // Launch add() kernel on GPU with N blocks
    add<<<N,1>>>(d_a, d_b, d_c);

    // Copy result back to host
    cudaMemcpy(c, d_c, size, cudaMemcpyDeviceToHost);

    // Cleanup
    free(a); free(b); free(c);
    cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);
    return 0;
}
```

CUDA Threads

- **Terminology**: a block can be split into parallel ***threads***

- Let's change `vector_addition()` to use parallel *threads* instead of parallel *blocks*

```
__global__ void add(int *a, int *b, int *c) {
    c[threadIdx.x] = a[threadIdx.x] + b[threadIdx.x];
}
```

- We use `threadIdx.x` instead of `blockIdx.x`

- Need to make one change in `main()`…

Vector Addition Using Threads: `main()`

```
#define N 512
int main(void) {
    int *a, *b, *c;             // host copies of a, b, c
    int *d_a, *d_b, *d_c;       // device copies of a, b, c
    int size = N * sizeof(int);

    // Alloc space for device copies of a, b, c
    cudaMalloc((void **)&d_a, size);
    cudaMalloc((void **)&d_b, size);
    cudaMalloc((void **)&d_c, size);

    // Alloc space for host copies of a, b, c
    //                              and setup input values
    a = (int *)malloc(size); random_ints(a, N);
    b = (int *)malloc(size); random_ints(b, N);
    c = (int *)malloc(size);
```

Vector Addition Using Threads: `main()`

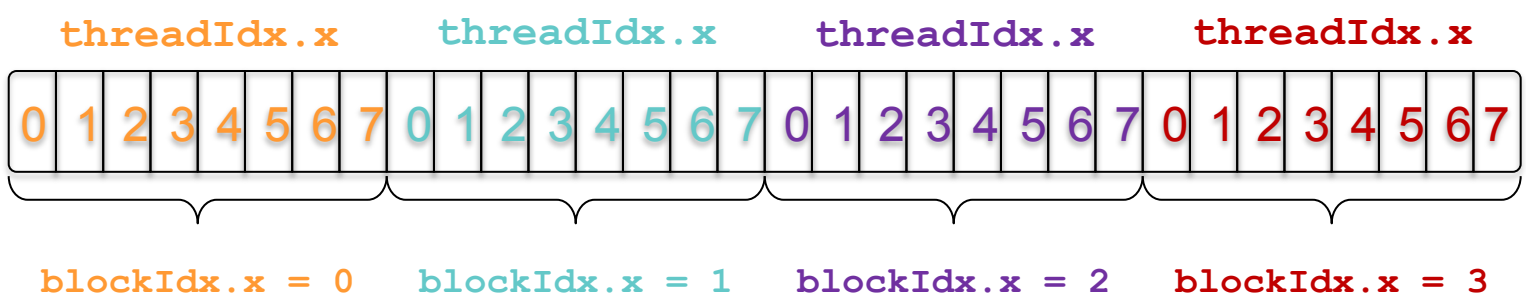
```
    // Copy inputs to device
    cudaMemcpy(d_a, a, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_b, b, size, cudaMemcpyHostToDevice);

    // Launch add() kernel on GPU with N threads
    add<<<1,N>>>(d_a, d_b, d_c);

    // Copy result back to host
    cudaMemcpy(c, d_c, size, cudaMemcpyDeviceToHost);

    // Cleanup
    free(a); free(b); free(c);
    cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);
    return 0;
}
```

Combining Blocks and Threads

- We've seen parallel vector addition using:
  - Many blocks with one thread each
  - One block with many threads

Indexing Arrays with Blocks and Threads

- No longer as simple as using `blockIdx.x` and `threadIdx.x`
  - Consider indexing an array with one element per thread (8 threads/block)

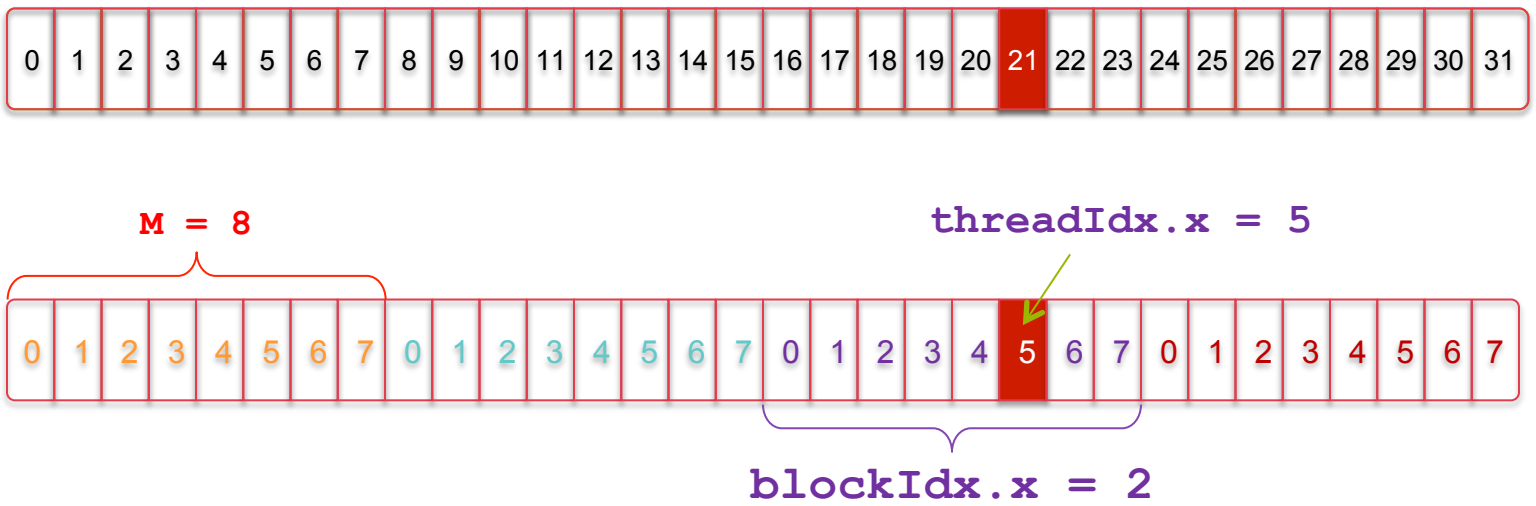


- With M threads/block a unique index for each thread is given by:

```
int index = threadIdx.x + blockIdx.x * M;
```

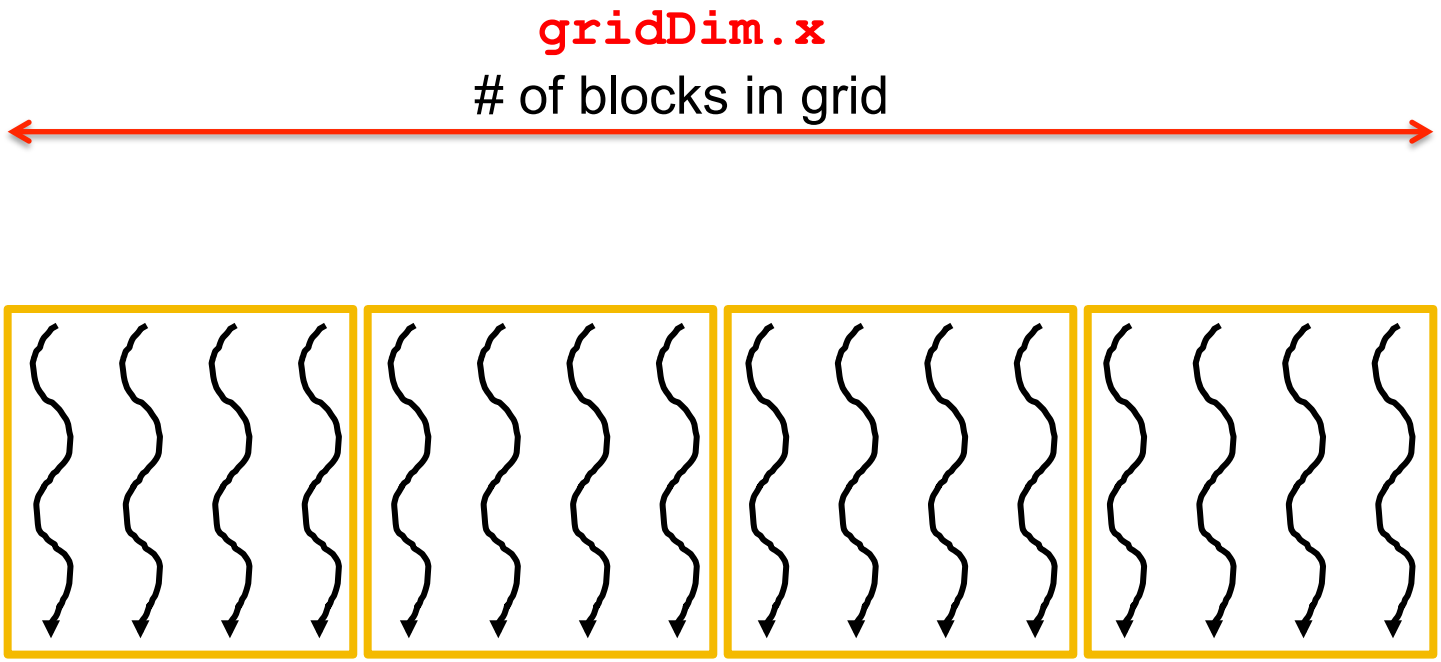
Indexing Arrays: Example

- Which thread will operate on the red element?

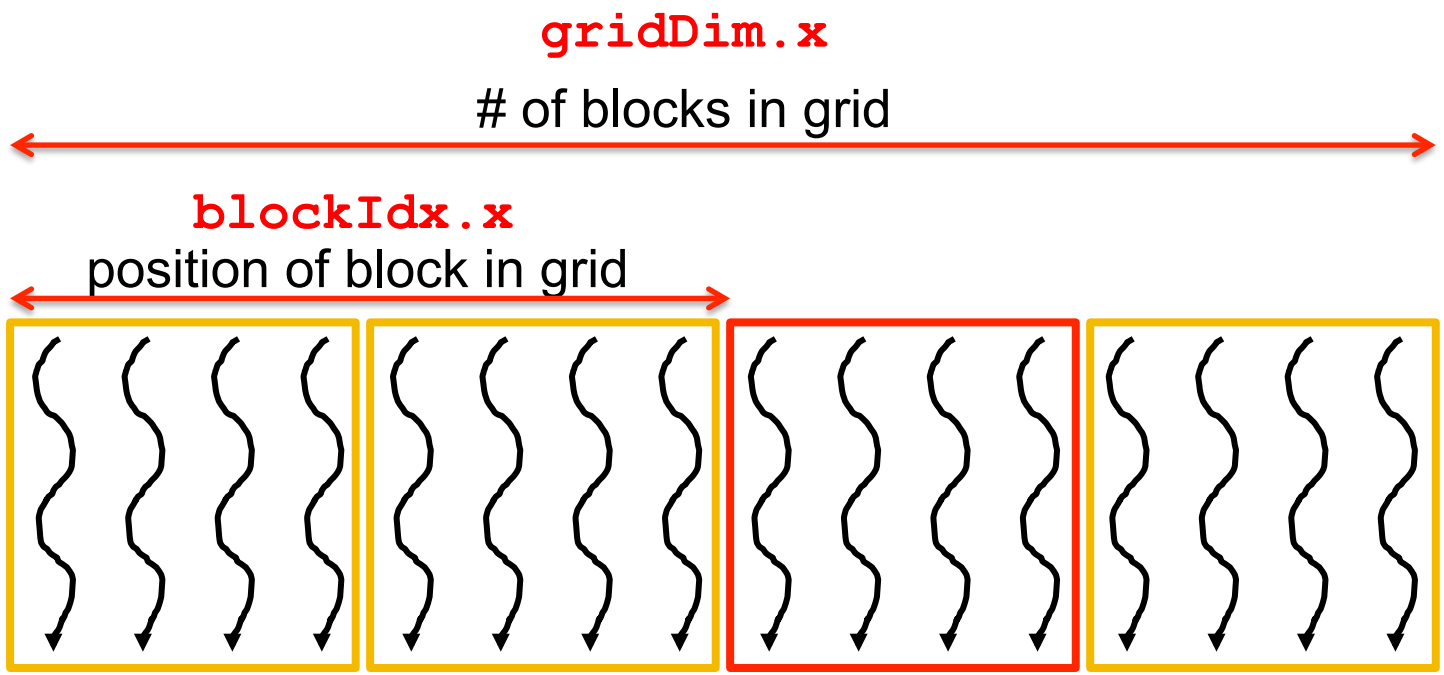



```
int index = threadIdx.x + blockIdx.x * M;
          =       5     +      2      * 8;
          = 21;
```

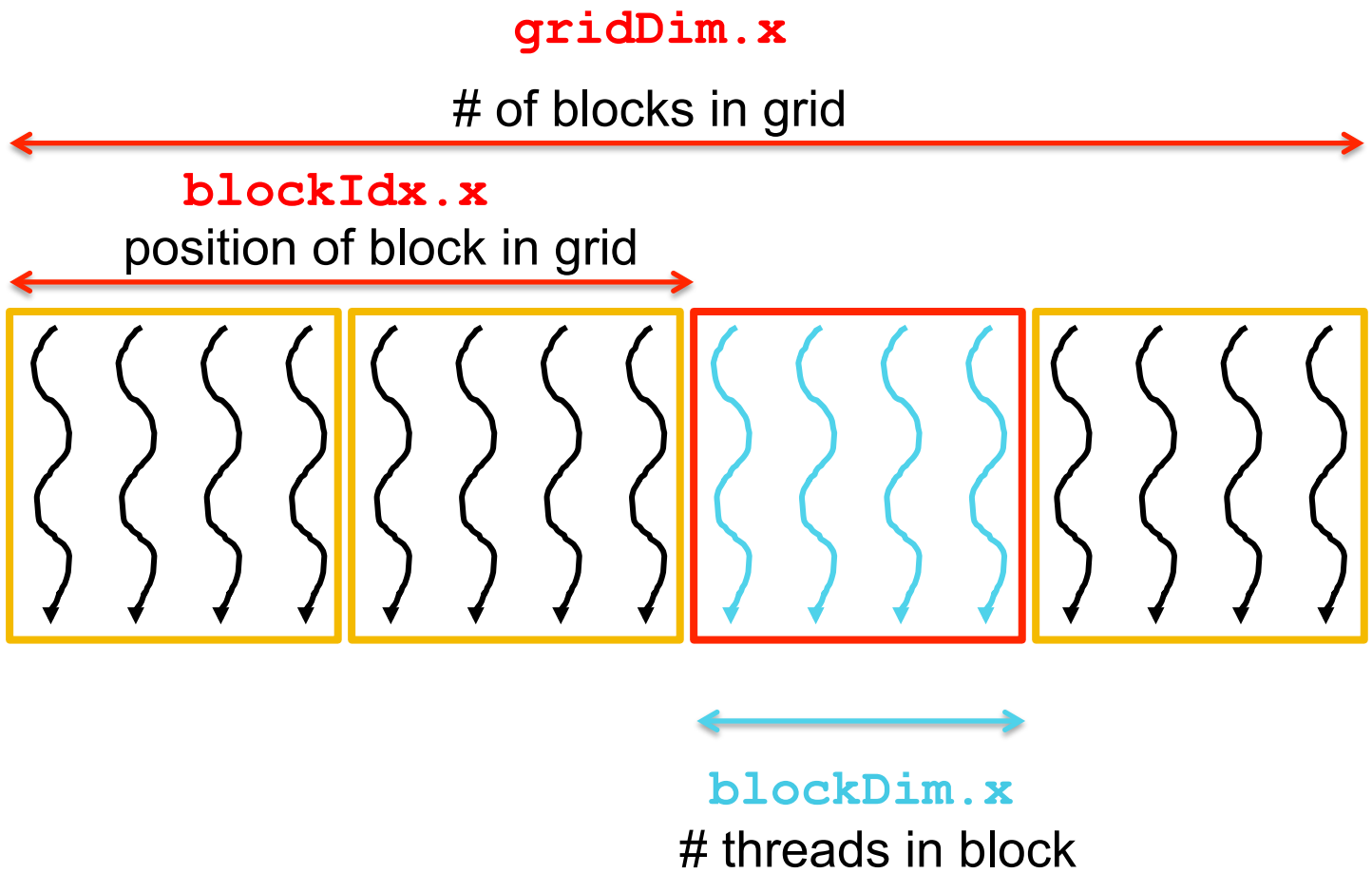
Grid Dimension



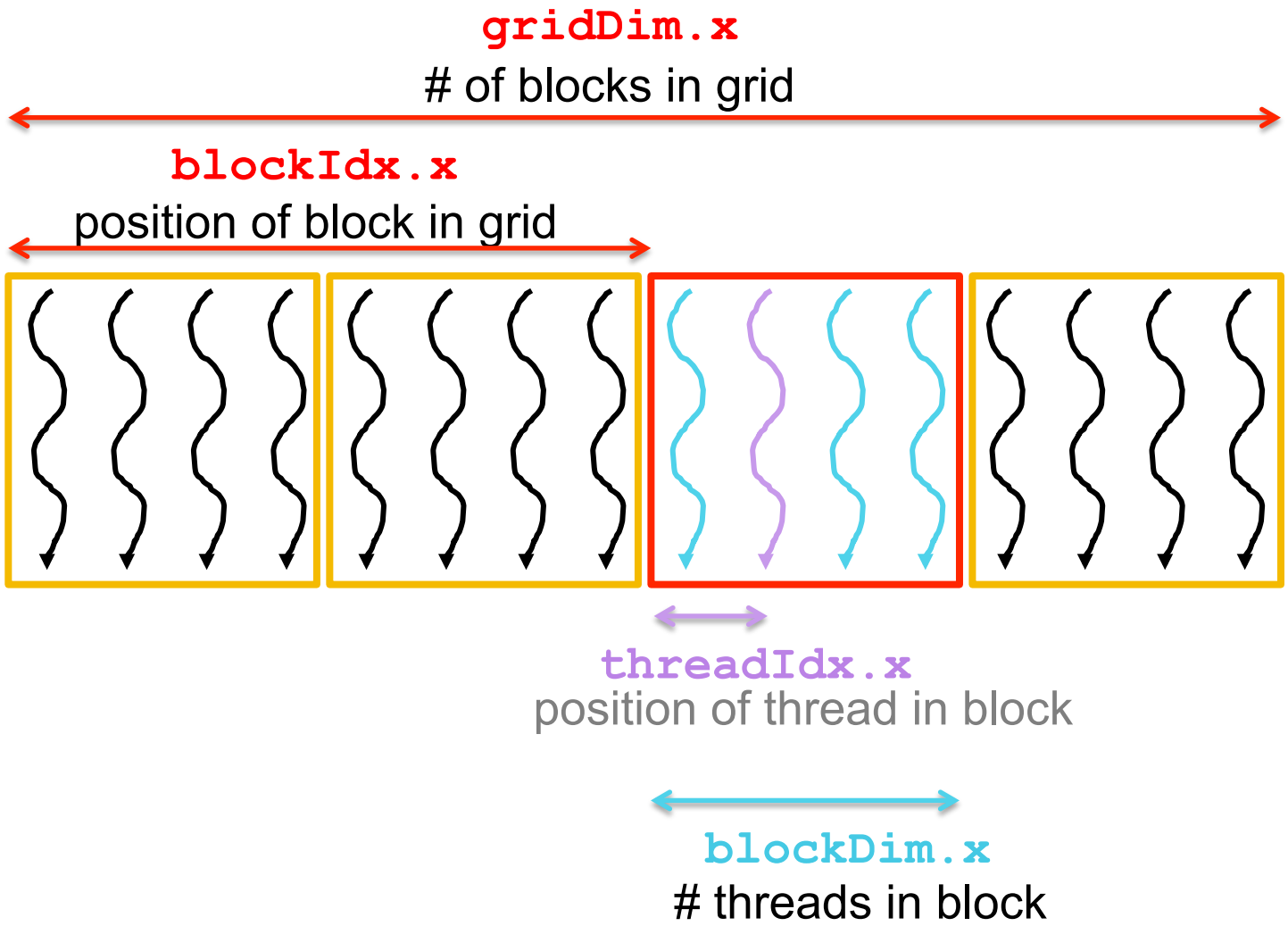
Block Index


gridDim.x
# of blocks in grid
blockIdx.x
position of block in grid

Block Dimension



Thread Index



Vector Addition with Blocks and Threads

- Use the built-in variable **`blockDim.x`** for threads per block

```
int index = threadIdx.x + blockIdx.x * blockDim.x;
```

- Combined version of `vector_addition()` to use parallel threads *and* parallel blocks

```
__global__ void add(int *a, int *b, int *c) {
    int index = threadIdx.x + blockIdx.x * blockDim.x;
    c[index] = a[index] + b[index];
}
```

- What changes need to be made in **`main()`**?

Addition with Blocks and Threads: `main()`

```
#define N (2048*2048)
#define THREADS_PER_BLOCK 512
int main(void) {
    int *a, *b, *c;           // host copies of a, b, c
    int *d_a, *d_b, *d_c;     // device copies of a, b, c
    int size = N * sizeof(int);

    // Alloc space for device copies of a, b, c
    cudaMalloc((void **)&d_a, size);
    cudaMalloc((void **)&d_b, size);
    cudaMalloc((void **)&d_c, size);

    // Alloc space for host copies of a, b, c
    //                            and setup input values
    a = (int *)malloc(size); random_ints(a, N);
    b = (int *)malloc(size); random_ints(b, N);
    c = (int *)malloc(size);
```

Addition with Blocks and Threads: `main()`

```
    // Copy inputs to device
    cudaMemcpy(d_a, a, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_b, b, size, cudaMemcpyHostToDevice);

    // Launch add() kernel on GPU
    add<<<N/THREADS_PER_BLOCK,THREADS_PER_BLOCK>>>(d_a, d_b, d_c);

    // Copy result back to host
    cudaMemcpy(c, d_c, size, cudaMemcpyDeviceToHost);

    // Cleanup
    free(a); free(b); free(c);
    cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);

    return 0;
}
```

Handling Arbitrary Vector Sizes

- Typical problems are not always multiples of `blockDim.x`
- Avoid accessing beyond the end of the arrays:

```
__global__ void add(int *a, int *b, int *c, int n) {
    int index = threadIdx.x + blockIdx.x * blockDim.x;
    if (index < n)
        c[index] = a[index] + b[index];
}
```

- Update the kernel launch:

```
add<<<(N + M-1) / M,M>>>(d_a, d_b, d_c, N);
```

COOPERATING THREADS

Why Bother with Threads within the blocks?

- Having threads within blocks seem unnecessary
  - They add a level of complexity
  - What do we gain?

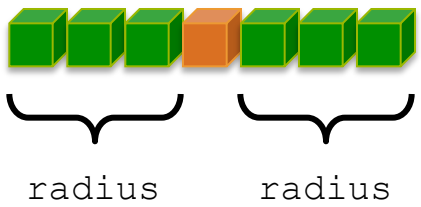
- Unlike parallel blocks, threads in the same block have mechanisms to:
  - Communicate
  - Synchronize

- To look closer, we need a new example…

1D Stencil

- Consider applying a 1D stencil to a 1D array of elements
  - Each output element is the sum of input elements within a radius

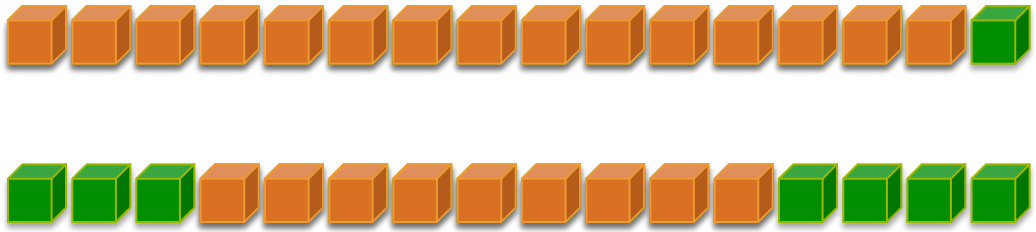
- If radius is 3, then each output element is the sum of 7 input elements:



Implementing Within a Block

- Each thread processes one output element
  - blockDim.x elements per block

- Input elements are read several times
  - With radius 3, each input element is read seven times

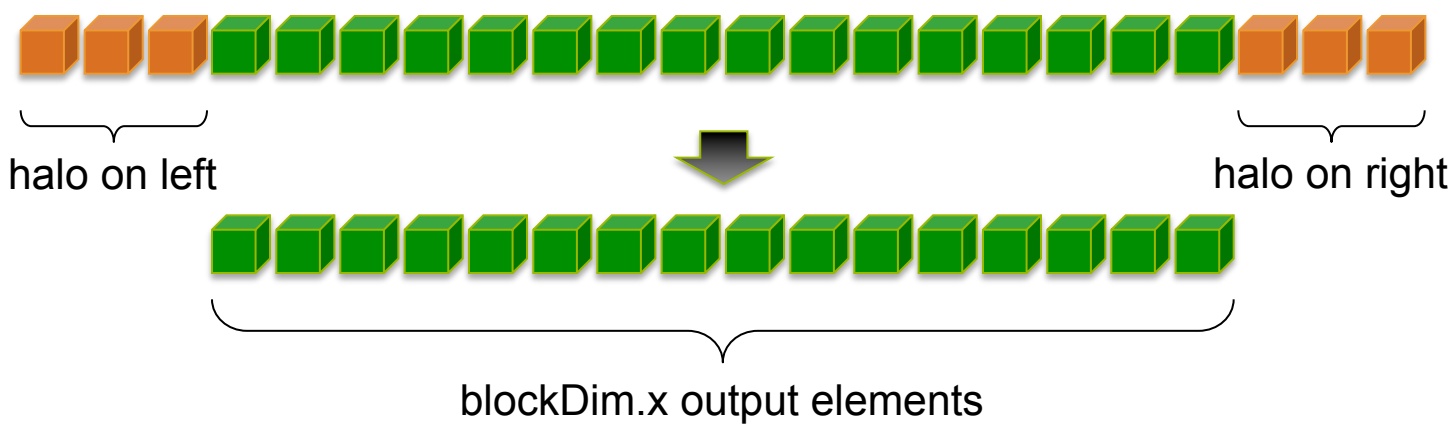


Sharing Data Between Threads

- Terminology: within a block, threads share data via shared memory
- Extremely fast on-chip memory, user-managed
- Declare using `__shared__`, allocated per block
- Data is not visible to threads in other blocks

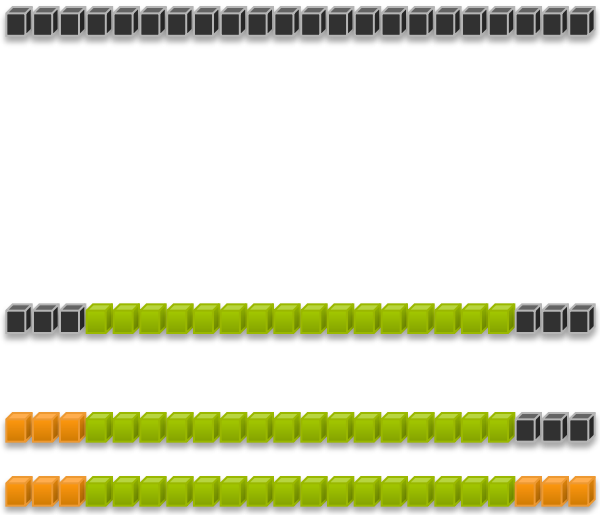
Implementing With Shared Memory

- Cache data in shared memory
  - Read (blockDim.x + 2 * radius) input elements from global memory to shared memory
  - Compute blockDim.x output elements
  - Write blockDim.x output elements to global memory
  - Each block needs a halo of radius elements at each boundary



Stencil Kernel

```
__global__ void stencil_1d(int *in, int *out) {
  __shared__ int temp[BLOCK_SIZE + 2 * RADIUS];
  int gindex = threadIdx.x + blockIdx.x * blockDim.x;
  int lindex = threadIdx.x + RADIUS;

  // Read input elements into shared memory
  temp[lindex] = in[gindex];
  if (threadIdx.x < RADIUS) {
    temp[lindex - RADIUS] = in[gindex - RADIUS];
    temp[lindex + BLOCK_SIZE] =
      in[gindex + BLOCK_SIZE];
  }
```



Stencil Kernel

```
  // Apply the stencil
  int result = 0;
  for (int offset = -RADIUS ; offset <= RADIUS ; offset++)
    result += temp[lindex + offset];

  // Store the result
  out[gindex] = result;
}
```

Data Race!

- The stencil example will not work…
- Suppose thread 15 reads the halo before thread 0 has fetched it…

Store at temp[18]

```
temp[lindex] = in[gindex];
if (threadIdx.x < RADIUS) {
  temp[lindex - RADIUS = in[gindex - RADIUS];
  temp[lindex + BLOCK_SIZE] = in[gindex + BLOCK_SIZE];
}
```

Skipped, threadIdx > RADIUS

Load from temp[19]

```
int result = 0;
result += temp[lindex + 1];
```

__syncthreads()

- `void __syncthreads();`

- Synchronizes all threads within a block
  - Used to prevent RAW / WAR / WAW hazards

- All threads must reach the barrier
  - In conditional code, the condition must be uniform across the block

Stencil Kernel

```
__global__ void stencil_1d(int *in, int *out) {
    __shared__ int temp[BLOCK_SIZE + 2 * RADIUS];
    int gindex = threadIdx.x + blockIdx.x * blockDim.x;
    int lindex = threadIdx.x + radius;

    // Read input elements into shared memory
    temp[lindex] = in[gindex];
    if (threadIdx.x < RADIUS) {
        temp[lindex - RADIUS] = in[gindex - RADIUS];
        temp[lindex + BLOCK_SIZE] = in[gindex + BLOCK_SIZE];
    }

    // Synchronize (ensure all the data is available)
    __syncthreads();
```

Stencil Kernel

```
    // Apply the stencil
    int result = 0;
    for (int offset = -RADIUS ; offset <= RADIUS ; offset++)
        result += temp[lindex + offset];

    // Store the result
    out[gindex] = result;
}
```

MANAGING THE DEVICE

Coordinating Host & Device

- Kernel launches are **asynchronous**
  - Control returns to the CPU immediately

- CPU needs to synchronize before consuming the results

`cudaMemcpy()`	Blocks the CPU until the copy is complete Copy begins when all preceding CUDA calls have completed
`cudaMemcpyAsync()`	Asynchronous, does not block the CPU
`cudaDeviceSynchronize()`	Blocks the CPU until all preceding CUDA calls have completed

Reporting Errors

- All CUDA API calls return an error code (`cudaError_t`)
  - Error in the API call itself

    OR
  - Error in an earlier asynchronous operation (e.g. kernel)

- Get the error code for the last error:

```
cudaError_t cudaGetLastError(void)
```

- Get a string to describe the error:

```
char *cudaGetErrorString(cudaError_t)
```

```
printf("%s\n", cudaGetErrorString(cudaGetLastError()));
```

Device Management

- Application can query and select GPUs

```
cudaGetDeviceCount(int *count)
cudaSetDevice(int device)
cudaGetDevice(int *device)
cudaGetDeviceProperties(cudaDeviceProp *prop, int device)
```

- Multiple threads can share a device

- A single thread can manage multiple devices

  `cudaSetDevice(i)` to select current device

  `cudaMemcpy(…)` for peer-to-peer copies †

† requires OS and device support

Source

- Slides adapted from multiple sources:
  - NVIDIA teaching kit
  - Wen-mei W. Hwu, David B. Kirk, and Izzat El Hajj. *Programming Massively Parallel Processors: A Hands-on Approach*. Morgan Kaufmann, 2022.
  - CUDA C++ Programming Guide: https://docs.nvidia.com/cuda/cuda-c-programming-guide